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W6-02: CAP and BASE

COMPSCI4064 (H) and COMPSCI5088 (M)

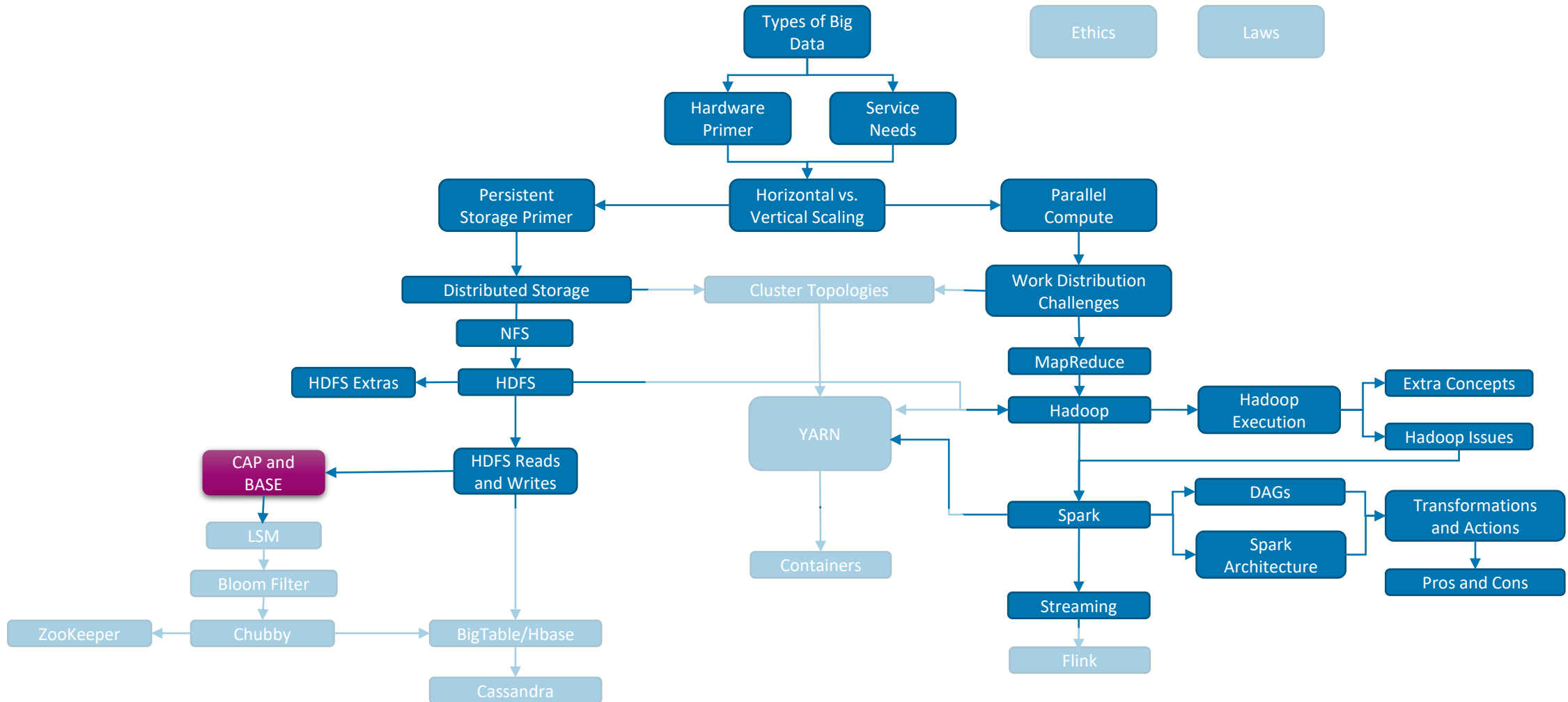
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Where are we?



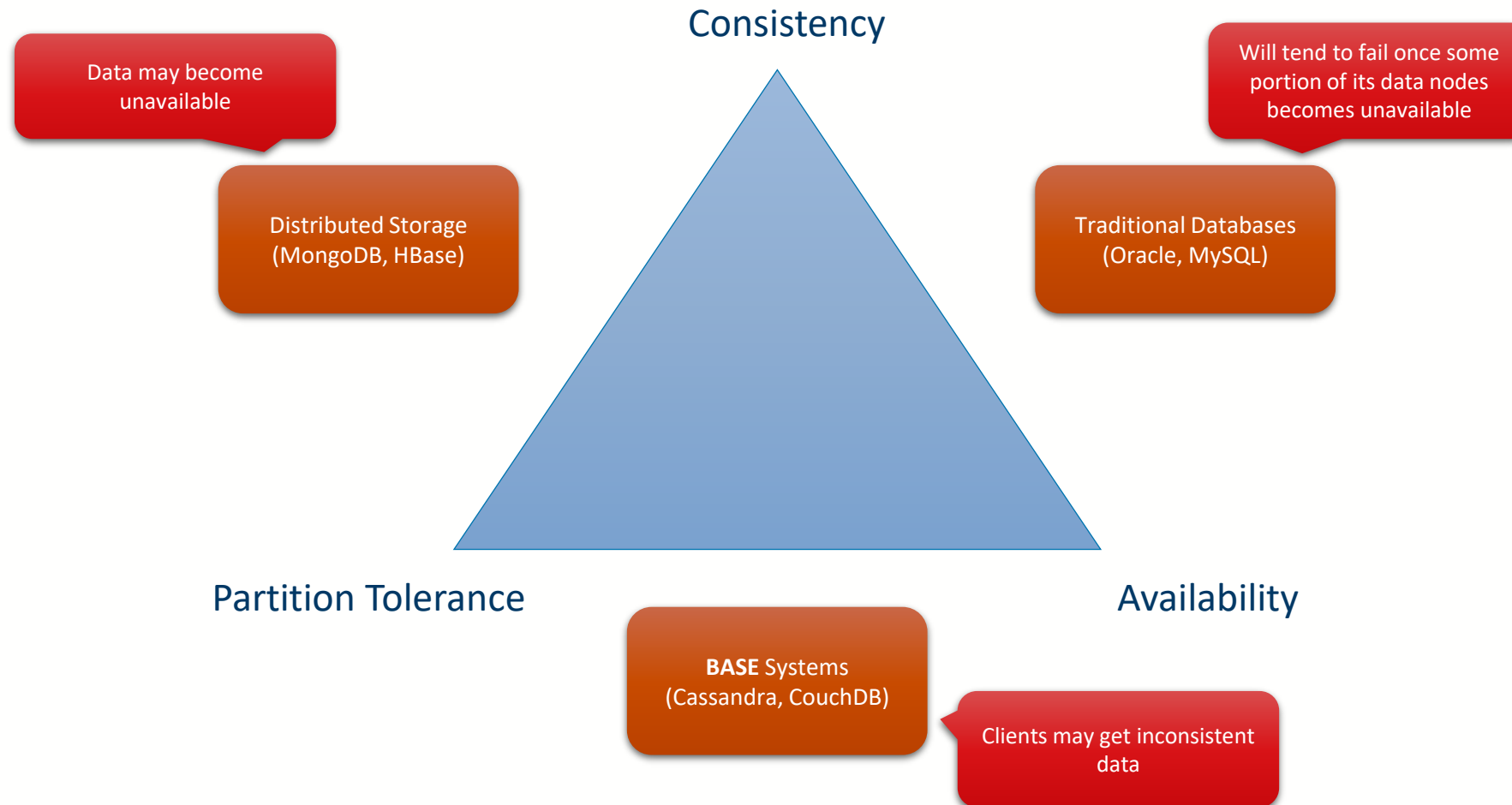


CAP Theorem

- CAP Theorem was developed by Eric Brewer in the beginning of the 2000's, and describes three desirable properties of a data storage system:
 - **Consistency**: Every node has the same knowledge of data at any instant of time
 - **Availability**: A guarantee that every request receives a response which may be processed or failed
 - **Partition Tolerance**: The system continues to operate even if a message is lost or part of the system fails
- In the majority of instances, **a distributed system can only guarantee two of the features**, not all three



CAP Examples





NFS and HDFS under CAP

NFS

- **Consistency:** **Yes** – there is only one copy of the data, so it is consistent
- **Availability:** **Decent** – it will always return data assuming the server is alive.
- **Partition Tolerance:** **No** – there is only one copy of the data, so a failure will bring the platform down

HDFS

- **Consistency:** **Mediocre** – the HDFS will make a best effort attempt to assure consistency of replicas, but this takes time
- **Availability:** **Decent?** – the HDFS will always return data so long as there is at least one available replica. What if the **Name Node goes down though?**
- **Partition Tolerance:** **Decent** – the HDFS is resilient to some degree of failures depending on replication factor



Big Data Systems and Consistency

- What you will find is that the more we push towards Big Data, the **easier availability and fault tolerance becomes**, but **consistency becomes more difficult**
 - This is because the more replicas of data or services we can support the more resilient we become
 - On the other hand, the more costly and difficult it becomes to ensure that all replicas are up-to-date and the more edge-cases you need to account for in the design (think HDFS recovery)



- Many modern Big Data Storage systems are what we would refer to as **BASE** systems:
 - **Basically Available**: The system guarantees that it will be available and will provide a response, but the response may be old data, or simply a notification that the desired data is unavailable
 - **Soft state**: The system's internal storage state will be constantly changing, even potentially when there is no new input
 - **Eventual consistency**: The system will eventually become consistent if new input is stopped being provided, but consistency is not guaranteed per transaction.
- They usually focus on high Availability and Partition Tolerance, at the expense of Consistency



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